HALF-SPIN SYMMETRY FOR CONORMAL CYCLES

Lemma 1. *There is a one-parameter semistable family of surfaces $\mathcal{S} \subset \mathcal{A} \to T$, in a smooth family of abelian fourfolds, with the following properties.*

(1) For $t \neq 0$ general, $S_t \subset A_t$ is a smooth irreducible Schoen surface and its inclusion is its Albanese embedding. After a finite base change and translation by a relative section which is zero at $t = 0$, the whole embedded family is invariant under relative inversion: $[-1]_{\mathcal{A}}(\mathcal{S}) = \mathcal{S}$.

(2) The central fibre is

$$S_0 = \Delta_J \cup (C \times C) \subset J \times J, \qquad \Delta_J \cap (C \times C) = \Delta_C,$$

where C is a smooth genus-two curve, embedded as a symmetric theta divisor in its Jacobian J.

(3) The projectivized conormal Gauss map of a general S_t has degree eight. Its quotient by inversion has degree four and is the incidence cover obtained by intersecting a line on a smooth quadric threefold with a quartic divisor on that quadric.

Proof. We use the following results from the literature, stated completely in the form needed below.

- **Schoen smoothing theorem.** The reducible surface obtained by gluing J and $C \times C$ along $C \subset J$ and $\Delta_C \subset C \times C$ deforms to a smooth surface of general type with
$$c_1^2 = 16 = 2c_2, \qquad q = 4, \qquad p_g = 5.$$
Its deformation space is locally smooth and irreducible of dimension four. This is Rito–Roulleau–Sarti, `paper_id=Rito_Roulleau_Sarti_Explicit_Schoen_surfaces`, `theorem_id=Theorem1`, `arXiv=1609.02235`; the smoothing and four-dimensional deformation assertions are also Theorem 2.2 of Ciliberto–Mendes Lopes–Roulleau, `paper_id=Ciliberto_MendesLopes_Roulleau_On_Schoen_surfaces`, `theorem_id=Theorem2.2`, `arXiv=1303.1750`.
- **Canonical-map theorem.** For a general Schoen surface S, the canonical map is a finite morphism of degree two onto a surface $Y \subset \mathbf{P}^4$ with forty nodes; Y is the complete intersection of a quadric and a quartic, and the canonical double cover is ramified exactly over the forty nodes. This is Ciliberto–Mendes Lopes–Roulleau, `paper_id=Ciliberto_MendesLopes_Roulleau_On_Schoen_surfaces`, `theorem_id=Theorem3.1`, `arXiv=1303.1750`.
- **Explicit-family theorem.** If $I_4 \subset \mathbf{P}^4$ is the Igusa quartic, there is a four-parameter family of quadrics $Q_{a,b,c,d}$ such that a general $I_4 \cap Q_{a,b,c,d}$ has exactly forty nodes, those nodes are two-divisible, and the double cover ramified at them is a Schoen surface. This is Rito–Roulleau–Sarti, `paper_id=Rito_Roulleau_Sarti_Explicit_Schoen_surfaces`, `theorem_id=Theorem3`, `arXiv=1609.02235`.
- **Albanese assertion.** A Schoen surface has irregularity four, hence a four-dimensional Albanese variety, and its Albanese map embeds it in that variety. This is the assertion recalled in the introduction of

paper_id=Rito_Roulleau_Sarti_Explicit_Schoen_surfaces, theorem_id=Introduction,Albanese-embeddingassertion, arXiv=1609.02235.

- **Limit Hodge-class theorem.** For a general genus-two Jacobian J, put $S_0 = \Delta_J + (C \times C)$. Its class lies in the unique three-dimensional Weil-Hodge space B_K^2 for $K = \mathbf{Q}(\sqrt{-3})$, with

$$\frac{-1+\sqrt{-3}}{2} \longmapsto \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix},$$

and S_0 deforms in the corresponding family of abelian fourfolds of Weil type. This is van Geemen, paper_id=van_Geemen_Weil_Classes_and_Decomposable_Abelian_Fourfolds, theorem_id=Section6.2,TheclassoftheSchoensurface, arXiv=2108.02087.

These statements give the family and its central fibre. The canonical deck involution has no invariant holomorphic one-forms: its quotient is the complete-intersection surface Y, which has irregularity zero. It therefore acts as -1 on the four-dimensional space of holomorphic one-forms of S_t.

We also make this translation coherently in the family. In Schoen's construction the canonical double covers, including their deck involutions, form a relative double cover; after shrinking and a finite base change, write the relative involution as ι. On the relative Albanese scheme its linear part is -1, so with respect to the given central Albanese embedding it acts as

$$x \longmapsto -x + c(t)$$

for a section $c : T \to \mathcal{A}$. The central involution is inversion on $\Delta_J \cup (C \times C)$, hence $c(0) = 0$. More intrinsically, even if one only takes the closure of the involutions from the punctured family, specialization gives $-S_0 + c(0) = S_0 = -S_0$, so translation by $c(0)$ stabilizes S_0. It must preserve the unique abelian-surface component Δ_J, forcing $c(0) = (u, u)$; it must then preserve $C \times C$, so $C + u = C$. A theta curve in a general genus-two Jacobian has trivial translation stabilizer, hence $u = 0$. Thus in either description $c(0) = 0$. Since multiplication by two on an abelian scheme is finite etale, after a further finite etale base change there is a section b through the origin with $2b = c$. Translating the relative Albanese embedding by $-b$ changes the displayed affine involution to $x \mapsto -x$. Since $b(0) = 0$, it leaves the embedded central fibre unchanged. This supplies the relative inversion pairing used below, not merely unrelated translations on individual fibres.

Let $V = H^0(S_t, \Omega^1_{S_t})^*$. The wedge map

$$\bigwedge^2 H^0(S_t, \Omega^1_{S_t}) \longrightarrow H^0(S_t, K_{S_t})$$

is onto and has a one-dimensional kernel generated by a rank-four alternating form. Thus the tangent planes of the Albanese embedding are Lagrangian for a symplectic form on V, and their tangent-plane Gauss map is the canonical map

$$S_t \longrightarrow Y_t \subset \mathrm{LG}(2, V) = Q^3 \subset \mathbf{P}^4.$$

For $[v] \in \mathbf{P}(V)$, the Lagrangian planes containing v form a line $L_v \subset Q^3$. Since Y_t is a quartic divisor on Q^3, a general L_v meets Y_t in four points. Each has two inverse lifts in S_t, so the conormal Gauss degree is eight. Equivalently,

$$\deg(\gamma_{S_t}) = \int_{\mathbf{P}(N^*_{S_t/A_t})} h^3 = \int_{S_t} c_2(\Omega^1_{S_t}) = c_2(S_t) = 8.$$

□

Lemma 2. *For a general member of the Schoen family in Lemma 1, the natural monodromy of the degree-eight conormal Gauss map is exactly*

$$W(D_4) = (\mathbf{Z}/2\mathbf{Z})^3 \rtimes \mathfrak{S}_4$$

on the four inverse pairs.

Proof. First we certify that the quotient incidence cover has full $\mathfrak{S}_4$-monodromy. Use the five-symmetric forty-nodal member of Rito–Roulleau–Sarti, `paper_id=Rito_Roulleau_Sarti_Explicit_Schoen_surfaces`, `theorem_id=Section4, thefive-symmetricmember`, `arXiv=1609.02235`. The complete result used from that section is that its canonical image is the intersection of the displayed smooth quadric and Igusa quartic, has exactly forty nodes, and its double cover ramified at those nodes is a Schoen surface.

Fourier coordinates put the quadric in the form $a_1a_4 + a_2a_3 - 3a_0^2 = 0$. Restrict the Igusa quartic to the pencil

$$(a_0, \dots, a_4) = (1, 1, 0, 0, 3) + s(t, 0, 3, t^2, 6t).$$

The restricted Igusa equation is $80f(s,t)$, where the full polynomial is

$$\begin{aligned} f(s,t) = {} & 18s^4t^6 + 216s^4t^5 + 9s^4t^4 + 324s^4t^3 + 162s^4t \\ & + s^3t^6 + 27s^3t^5 + 324s^3t^4 + 90s^3t^3 + 648s^3t^2 + 27s^3t + 81s^3 \\ & + 9s^2t^4 + 165s^2t^3 + 81s^2t^2 + 405s^2t + 27s^2 \\ & + 30st^2 + 36st + 84s + 9. \end{aligned} \tag{6}$$

It has specialization

$$f(s,1) = 729s^4 + 1198s^3 + 687s^2 + 150s + 9,$$

which is irreducible modulo 29. Its cubic resolvent specializes to $z^3 - 687z^2 + 153456z - 11289708$, whose reduction $z^3 - 2z^2 + z + 2$ has no root modulo 5. Its discriminant is a nonzero exactly $-27D(t)$, where:

(7)

$$\begin{aligned} D(t) = {} & 733t^{24} + 38952t^{23} + 514710t^{22} - 1970352t^{21} + 9891645t^{20} - 48716532t^{19} \\ & + 219511216t^{18} - 950504112t^{17} + 3112479432t^{16} - 8675155404t^{15} + 18869978862t^{14} \\ & - 36772101360t^{13} + 60600367404t^{12} - 99990794736t^{11} + 146694965598t^{10} \\ & - 216718347636t^9 + 272773942416t^8 - 328440510720t^7 + 347712162264t^6 \\ & - 335107405260t^5 + 320719506237t^4 - 262644440760t^3 + 247999068414t^2 - 152128174896t + 418419442 \end{aligned}$$

For a reproducible squarefreeness certificate, reduction modulo 17 factors as

$$\begin{aligned} D(t) = {} & 2(t+7)(t^3 + 4t^2 - 8t - 1)(t^6 - 6t^4 - 8t^3 - 7t^2 + 6t + 6) \\ & \cdot (t^7 + 8t^6 + 8t^5 + 7t^4 - 8t^3 + 6t^2 - 4t - 4) \\ & \cdot (t^7 - 8t^6 - 4t^4 - 8t^3 - 3t^2 - 2t - 6) \pmod{17}. \end{aligned} \tag{8}$$

The five displayed factors are pairwise coprime and each is coprime to its derivative (ordinary Euclidean division in $\mathbf{F}_{17}[t]$); equivalently $\gcd(D, D') = 1$ modulo 17. This is an exact, directly reproducible certificate, not a numerical root computation. Hence the generic quartic and its cubic resolvent are irreducible and its discriminant is not a square. The quartic Galois criterion gives arithmetic group $\mathfrak{S}_4$. A simple

The Euclidean algorithm also gives $\gcd(D, \mathrm{lc}_s(f)) = 1$ in $\mathbf{Q}[t]$. Hence every simple zero of D is away from the leading-coefficient degeneration and gives a quartic with exactly one ordinary double root, so its local geometric inertia is a transposition. The geometric group is normal in the arithmetic group, so it too is $\mathfrak{S}_4$. The restricted pencil already has full group, hence so does the global incidence cover of this explicit Schoen member.

Let $G \subset W(B_4)$ be the monodromy on the eight lifts. We show $G \subset W(D_4)$. On $\mathbf{P}\Lambda_{S_t}$, inversion can have fixed points only above $S_t \cap A_t[2]$. Each corresponding projective conormal fibre has dimension one, so this fixed locus has codimension two in the threefold $\mathbf{P}\Lambda_{S_t}$.

We spell out the local-inertia implication needed here. Let D be a branch divisor in the Gauss base, and let $g \in W(B_4)$ generate the tame inertia at its generic point. Write the signed permutation g as signed cycles on the four inverse pairs. A cycle is *negative* when going once around it carries a chosen sheet to the inverse sheet. The product-sign character satisfies

$$\chi(g) = (-1)^{\#\{\text{negative signed cycles of } g\}}. \tag{0}$$

If $\chi(g) = -1$, choose a sheet in one negative cycle, of length m on the inverse pairs. In the normalization over a transverse disk to D, the corresponding $2m$ signed sheets form one inertia orbit and coalesce at one codimension-one point; the inverse of the chosen sheet lies in that same orbit. The limiting point is therefore fixed by inversion. As the generic point of D varies, these points form a fixed divisor in the normalization of $\mathbf{P}^3$ in the Gauss function field. The proper generically finite Gauss map factors through that normalization; since the source is smooth, the resulting proper birational morphism is an isomorphism at the generic point of every divisor on the normalization. The strict transform is therefore a fixed divisor in $\mathbf{P}\Lambda_{S_t}$. Thus odd product-sign inertia forces a divisorial inversion-fixed locus. Since the actual fixed locus has codimension two, $\chi(g) = 1$ for every divisorial inertia group.

Consequently the quadratic extension defined by χ is unramified at every codimension-one point of $\mathbf{P}^3$. Purity of the branch locus extends it to an etale double cover of $\mathbf{P}^3$, which is trivial because $\mathbf{P}^3(\mathbf{C})$ is simply connected. Thus every monodromy element changes an even number of signs.

We finally compute the actual incidence map at a node; the abstract A_1-cover alone would not suffice. In the explicit five-symmetric member, use affine coordinates $x_0, \dots, x_4$, put $S = \sum x_i$, and write its quadric and quartic as

$$Q = 5\sum x_i^2 - 7S^2, \qquad F = 4\left(\sum x_i^4 + S^4\right) - \left(\sum x_i^2 + S^2\right)^2. \tag{1}$$

The point

$$y = (3/5, 3/5, 3/5, -2/5, -2/5)$$

satisfies $Q(y) = F(y) = 0$, and direct differentiation gives

$$dQ_y = (-8, -8, -8, -18, -18), \qquad dF_y = (512/125, 512/125, 512/125, 1152/125, 1152/125) = -\frac{64}{125} dQ_y.$$

Thus the quadratic tangent cone of $Y = (Q = F = 0)$ inside the smooth quadric at y is induced by the Hessian of $F + (64/125)Q$. On $T_y Q/\mathbf{C}y$, take the images of

$$(-1, 1, 0, 0, 0), \quad (-1, 0, 1, 0, 0), \quad (-9/4, 0, 0, 1, 0).$$

In this basis, the quadratic form q_0 whose zero conic parametrizes the directions of lines on Q^3 through y, and the quadratic tangent cone q_Y, have matrices

$$[q_0] = \begin{pmatrix} 10 & 5 & 45/4 \\ 5 & 10 & 45/4 \\ 45/4 & 45/4 & 155/8 \end{pmatrix}, \qquad [q_Y] = \begin{pmatrix} 64/5 & 32/5 & 72/5 \\ 32/5 & 64/5 & 72/5 \\ 72/5 & 72/5 & 112/5 \end{pmatrix}. \tag{2}$$

Their determinants are $375/2$ and $12288/125$, respectively, so this is an ordinary node. More importantly,

$$[q_Y] - \frac{32}{25}[q_0] = \operatorname{diag}(0, 0, -12/5), \tag{3}$$

so the two conics are not equal. Therefore some line direction k through y has $q_0(k) = 0$ but $q_Y(k) \neq 0$.

Here is why this is exactly the required transversality for the conormal incidence map. Identify the smooth quadric with $\mathrm{LG}(2, V)$. Around the plane represented by y, choose a Lagrangian splitting $V = E \oplus E^*$. Nearby planes are graphs of symmetric matrices

$$M = \begin{pmatrix} x & z \\ z & y' \end{pmatrix}.$$

In the chart $[u] = [1 : t] \in \mathbf{P}(E)$, the universal incidence map is

$$(M, t) \longmapsto [1 : t : x + zt : z + y't]. \tag{4}$$

At fixed t, its transverse linear projection has kernel $(x, y', z) = (t^2, 1, -t)$. As t varies, these kernels trace precisely the rank-one conic $xy' - z^2 = 0$, which is the conic $q_0 = 0$ in (2). Choose $t = t_0$ corresponding to the direction k found above. Then the kernel of the actual incidence projection is nonisotropic for q_Y.

The holomorphic Morse lemma, applied after splitting off this nonisotropic kernel and with t as a parameter, now puts the quotient incidence map in the form

$$(t, w, A, B), \quad w^2 = AB, \qquad (t, w, A, B) \longmapsto (t, A, B).$$

The canonical double cover of the A_1-germ therefore makes the *actual signed conormal incidence map* analytically

$$(t, a, b) \longmapsto (t, A, B) = (t, a^2, b^2), \qquad (a, b) \sim (-a, -b) \tag{5}$$

on the quotient. A meridian around $A = 0$ reverses a, and one around $B = 0$ reverses b. Their product fixes the two quotient sheets but applies inversion on both signed pairs. Hence $\ker(G \to \mathfrak{S}_4)$ contains $e_i + e_j$. Conjugating by the established $\mathfrak{S}_4$-quotient gives all double sign changes, which generate $(\mathbf{Z}/2)^3$. Therefore $G = W(D_4)$.

We finish with a fibrewise generic-monodromy argument, including the signed kernel. Let M be the irreducible four-dimensional Schoen parameter space, shrunk so the surfaces and their conormal cycles form families, and let $\mathscr{D} \subset \mathbf{P}^3 \times M$ be the relative Gauss discriminant. Over

$$U_M = (\mathbf{P}^3 \times M) \setminus \mathscr{D}$$

form the oriented principal labeling cover $P_M \to U_M$ exactly as in the next lemma (after a finite etale base change on M, if necessary). Its degree is $|W(D_4)| = 192$. The inertia argument above applies fibrewise, so every fibre group is contained in $W(D_4)$.

At the explicit five-symmetric point m_0, equations (5)–(8) show that the fibre group is all of $W(D_4)$. Choose a base point $\xi_0 \in U_{m_0}$ and finitely many based loops $\gamma_1, \dots, \gamma_N \subset U_{m_0}$ whose signed permutations generate $W(D_4)$. Their union K is compact and $K \times \{m_0\}$ is disjoint from the closed set $\mathscr{D}$. Hence, after shrinking to a small Euclidean ball $V \subset M(\mathbf{C})$ about m_0, one has $K \times V \subset U_M$. The finite covering $P_M|_{K\times V} \to K \times V$ is topologically locally trivial. After labeling its fibre over (ξ_0, m_0), the monodromy permutations of each fixed loop $\gamma_i \times \{m\}$ are locally constant in m. They are therefore the same generators for every $m \in V$. Thus every fibre over V has monodromy $W(D_4)$.

It remains only to pass from this Euclidean neighbourhood to an algebraic generic statement. A fibre $(P_M)_m$ is geometrically connected exactly when the fibre monodromy equals $W(D_4)$: a subgroup $H \subset W(D_4)$ acts on the principal label set $W(D_4)$ by left multiplication, and this action is transitive precisely when $H = W(D_4)$. For the finite-type morphism $P_M \to M$, the locus where the geometric fibre is connected is constructible (the constructibility theorem for the number of geometric connected components of fibres). Denote it by M^{full}. It contains the nonempty Euclidean open V, which is Zariski dense because M is irreducible. A dense constructible subset of an irreducible variety contains a nonempty Zariski-open subset; that open is automatically dense. After replacing M^{full} by this open, every member in it has exact $W(D_4)$ fibre monodromy.

If $o \in \overline{M}$ is the fixed Schoen boundary point, density gives $o \in \overline{M^{\mathrm{full}}}$. A general algebraic curve through o is not contained in its proper closed complement; its normalization therefore supplies a one-parameter smoothing whose general fibre has the required full signed monodromy.

□

Lemma 3. *Choose a one-parameter Schoen smoothing as in Lemma 1 whose very general member has the monodromy of Lemma 2, and make a finite base change which labels the two parity cycles. On $A_0 = J \times J$, put*

$$Y_+ = \{(x+y, y) : x, y \in C\}, \qquad Y_- = \{(x, x+y) : x, y \in C\}.$$

Then the specializations of the middle Segre classes are

$$\begin{aligned} \operatorname{sp} s_2(\Lambda_+) &= 16\big([Y_+] + [\{0\} \times J]\big), \\ \operatorname{sp} s_2(\Lambda_-) &= 16\big([Y_-] + [J \times \{0\}]\big). \end{aligned} \tag{1}$$

Proof. We separate the part determined by the normalized labeled cover from the possible excess part and determine the latter without assuming that an unlabeled characteristic-cycle limit remembers a labeling.

The relative normalization on the finite locus

We use the following specialization theorem. Its complete statement is: if $W \to T$ is a smooth morphism to a smooth curve over a field of characteristic zero and $X \subset W$ is flat over T, then the scheme-theoretic fibre of the relative conormal cycle is an effective conic Lagrangian cycle; specialization is the Gysin map on Chow groups, sends effective cycles to effective cycles, and preserves the total conormal degree. This is Codogni–Kramer, `paper_id=Codogni_Kramer_Semicontinuity_of_Gauss_maps`, `theorem_id=Lemma2.3andProposition2.4`, `arXiv=2010.00053`.

For a simple-normal-crossing smoothing $X_1 \cup_D X_2$, the nearby constructible function is $\mathbf{1}_{X_1} + \mathbf{1}_{X_2} - 2\mathbf{1}_D$. With $CC(\mathbf{1}_M) = (-1)^{\dim M}\Lambda_M$, this gives in the

present family

$$\mathrm{sp}(\Lambda_{S_t}) = \Lambda_{C\times C} + 2\Lambda_{\Delta_C} + \Lambda_{\Delta_J}. \tag{2}$$

Only (2) as an unlabeled Lagrangian cycle is used here.

Write a cotangent direction on $J \times J$ as $[\alpha, \beta]$, and set

$$L_1 = \{\beta = 0\}, \qquad L_2 = \{\alpha = 0\}, \qquad L_3 = \{\alpha + \beta = 0\} \subset \mathbf{P}^3. \tag{3}$$

On $B^\circ = \mathbf{P}^3 \setminus (L_1 \cup L_2 \cup L_3)$, every component in (2) has finite Gauss fibre; Λ_{Δ_J} has no point there. We now construct proper relative labeled covers over the *whole* projective cotangent base rather than infer them from (2). Put $\mathcal{B} = \mathbf{P}^3 \times T$, and let

$$\Gamma \subset \mathcal{A} \times_T \mathcal{B}$$

be the scheme-theoretic closure of the projectivized conormal over the punctured parameter curve. It is proper over $\mathcal{B}$ and carries the relative inversion constructed in Lemma 1. Over a dense open $U^* \subset \mathcal{B} \cap \{t \neq 0\}$, it is the etale degree-eight cover with fixed-point-free inversion.

Over U^*, form inside $\Gamma^4_{U^*}$ the configuration cover of ordered tuples (p_1, p_2, p_3, p_4) containing one representative of each inversion pair. It has two monodromy orbits, distinguished by the orientation (the product of the four sign changes). Choose one orbit P^*. Exact $W(D_4)$-monodromy says that

$$P^* \longrightarrow U^*$$

is the connected principal $G = W(D_4)$-labeling cover, of degree $|G| = 192$. After the finite base change already allowed in the statement, this choice persists around the puncture.

Take the schematic closure of P^* in the proper space $\Gamma^4_{\mathcal{B}}$, and let $\overline{P}$ be its normalization. Then $\overline{P} \to \mathcal{B}$ is proper. It is integral and dominates T, so the parameter is a nonzerodivisor; over the parameter DVR this is equivalent to flatness. Every deck transformation of P^*/U^* extends uniquely to $\overline{P}$, because the latter is the normalization in its function field.

For a sign vector ε, let $G_\varepsilon \subset G$ be its stabilizer, of order 24, and put

$$\overline{H}_\varepsilon = \overline{P}/G_\varepsilon.$$

This quotient is integral, proper over $\mathcal{B}$, and flat over T; on U^* it is the associated degree-eight sign-choice cover. The four coordinate maps $p_j : \overline{P} \to \Gamma \to \mathcal{A}$ are regular. The group-law map

$$\sum_{j=1}^{4} \varepsilon_j p_j : \overline{P} \longrightarrow \mathcal{A}$$

is G_ε-invariant and descends to $q_\varepsilon : \overline{H}_\varepsilon \to \mathcal{A}$. Choose $\varepsilon_+ = (1,1,1,1)$ and $\varepsilon_- = (-1,1,1,1)$, and define global relative cycles by

$$\mathcal{L}_\pm = (q_{\varepsilon_\pm}, \gamma)_*[\overline{H}_{\varepsilon_\pm}] \quad \text{on } \mathcal{A} \times_T \mathcal{B}. \tag{4a}$$

On U^*, (4a) is exactly the definition in the problem: every parity sign choice occurs once, and if its summation map has generic degree e onto an image component, proper pushforward gives coefficient e. Thus (4a) proves retention of the generic summation multiplicities. Since the sources are flat over the smooth parameter curve, refined Gysin specialization commutes with these proper pushforwards. It also commutes with intersection by the relative hyperplane class h; therefore (4a)

specializes both the global cycles and all their Segre classes. In particular, intersection with h^3 shows that their total Gauss degree remains eight.

We also verify here that these special cycles are conic Lagrangian; this does not follow merely from conormal specialization. Work first over the smooth finite-etale locus and lift the projective covector to a local nonzero covector ξ. The four labeled conormal branches are maps $p_j = p_j(\xi)$ satisfying

$$d\xi \wedge dp_j = 0,$$

because the graph $(p_j(\xi), \xi) \subset T^*A$ is Lagrangian. For the signed sum $q_\varepsilon = \sum \varepsilon_j p_j$, the addition map gives

$$d\xi \wedge dq_\varepsilon = \sum_{j=1}^{4} \varepsilon_j (d\xi \wedge dp_j) = 0. \tag{4b}$$

Thus every generic parity-sum image is isotropic; it is conic and has the projective dimension three, hence its conic lift has dimension four and is Lagrangian. Equation (4b) is algebraic (vanishing of the pullback of the relative cotangent symplectic form), so it persists on the closure. Flatness over T makes every component of the special projective source three-dimensional. A component whose image has smaller dimension contributes zero to the proper pushforward cycle; every component which survives has three-dimensional projective image, and (4b) makes its four-dimensional conic lift isotropic, hence Lagrangian. Therefore the special fibres of $\mathcal{L}_\pm$, and every effective residual cycle obtained from them, are pure conic Lagrangian cycles. In particular each irreducible residual component is the conormal cycle of its support, so the conormal classification used below is legitimate.

We next restrict this global construction to the finite locus. Properness and upper semicontinuity of fibre dimension give an open neighbourhood $N \subset \mathcal{B}$ of $B^\circ \times \{0\}$ on which $\Gamma_N \to N$ is finite. The closure of P^* inside $(\Gamma_N)^4$, its normalization, and both stabilizer quotients are then finite over N; by uniqueness of normalization these are precisely the restrictions of $\overline{P}$ and $\overline{H}_{\varepsilon_\pm}$. Every component of their special fibres has dimension three and maps onto the three-dimensional N_0. Hence no component supported over a proper subset of B° is introduced there.

It remains to determine the two coincident pairs contributed by $2\Lambda_{\Delta_C}$. This is exactly where the previous unlabeled argument was insufficient, so we normalize locally. At a general point of Δ_C, an analytic change of coordinates puts the embedded smoothing in the form

$$w = 0, \qquad uv = t \tag{4}$$

inside coordinates (z, u, v, w) on the abelian fourfold, with z along Δ_C. Its conormal is spanned by dw and $v\,du + u\,dv$. For a fixed general cotangent direction whose du, dv coefficients are both nonzero, the conormal equations give $u/v = \kappa^2$ for a unit κ. After the normalization $t = s^2$, the two branches are

$$(u, v) = (s\kappa, s\kappa^{-1}), \qquad (u, v) = (-s\kappa, -s\kappa^{-1}). \tag{5}$$

Thus they are two correlated labeled sheets, not two formal copies of an unlabeled cycle. Both approach the same point of Δ_C, and inversion pairs the two branches near that point with the corresponding branches near its inverse. Since $\overline{P}$ is precisely the normalization of the ordered configuration closure, the two branches in (5) are distinct labeled sections there; call their inverse pairs c, d. The four branches from $C \times C$ are unramified at a general direction and give the two pairs a, b. Thus

this is a statement about the special fibre of the relative labeling cover constructed above, not only about the support of (2). It proves that representatives of the four inverse pairs may be chosen so that, as $s \to 0$,

$$a \to (p, q), \quad b \to (p, -q), \quad c \to (r, r), \quad d \to (r, r), \tag{6}$$

where $\{\pm p\}, \{\pm q\}, \{\pm r\}$ are the zero divisors on C of $\alpha, \beta, \alpha + \beta$, respectively. Formula (5) is the required normalization control of the multiplicity-two diagonal sheet.

For four signed pairs, representatives of the four inverse pairs of even half-spin sums are

$$\begin{aligned} &a + b + c + d, \quad a + b - c - d, \\ &a - b + c - d, \quad a - b - c + d, \end{aligned} \tag{7}$$

and for odd parity they are

$$\begin{aligned} &a + b + c - d, \quad a + b - c + d, \\ &a - b + c + d, \quad -a + b + c + d. \end{aligned} \tag{8}$$

Substitution of (6) into (7) gives the four points in the conormal fibre of $[2]Y_+$, together with the two points in the conormal fibre of $\{0\} \times [2]C$, each of the latter occurring twice. Substitution into (8) gives $[2]Y_-$ and twice the conormal of $[2]C \times \{0\}$. Hence the special parity cycles contain, with the displayed generic multiplicities,

$$\begin{aligned} D_+ &= [2]_* \Lambda_{Y_+} + 2[2]_* \Lambda_{\{0\} \times C}, \\ D_- &= [2]_* \Lambda_{Y_-} + 2[2]_* \Lambda_{C \times \{0\}}. \end{aligned} \tag{9}$$

These really are conormal, rather than only pointwise, limits: at $(p + r, r) \in Y_+$, the covector (α, β) kills the tangent because $\alpha(T_p C) = 0$ and $(\alpha + \beta)(T_r C) = 0$; at $(r, q + r) \in Y_-$, it kills the tangent because $(\alpha + \beta)(T_r C) = 0$ and $\beta(T_q C) = 0$. The coordinate-curve terms are checked in the same way. The first summand in either line has Gauss degree four and the second has Gauss degree $2 \cdot 2 = 4$. Since each parity cycle has Gauss degree eight and degree is preserved by specialization, (9) exhausts every component which dominates $\mathbf{P}^3$.

Confinement of every excess contribution

Let $R_\pm$ be the effective residual cycle after subtracting the components (9). The finiteness argument over B°, together with degree exhaustion, shows

$$\gamma(\operatorname{Supp} R_\pm) \subset L_1 \cup L_2 \cup L_3. \tag{10}$$

Thus (10) excludes both additional dominant surfaces and components over ordinary branch divisors; only the three genuinely non-finite cotangent lines remain.

We now classify everything in (10) that can contribute to s_2. Let L be one of the three lines and let $B_L \subset J \times J$ be the abelian surface with tangent space

$$T_0 B_L = \bigcap_{[\xi] \in L} \ker(\xi).$$

If a conormal component over L has surface support, its projective conormal fibre is itself a line, hence equals L at a general point; its tangent space is constantly $T_0 B_L$, so the support is a translate of B_L. If it has divisor support, every general tangent hyperplane still contains $T_0 B_L$; the divisor is invariant under B_L, and cutting its Gauss line by one hyperplane produces a nonnegative integral sum of translates of B_L. Components with smaller support do not contribute to s_2. It follows that the

entire residual contribution to s_2 is a nonnegative integral combination of the three classes $[B_{L_i}]$.

For (3), these surfaces are

$$B_{L_1} = \{0\} \times J, \qquad B_{L_2} = J \times \{0\}, \qquad B_{L_3} = \Delta_J. \tag{11}$$

This also controls the possible excess over the Gauss line of Δ_J: it can only contribute an integral multiple of $[\Delta_J]$; no unlisted surface class can occur.

Determination of the three multiplicities

It remains to determine the coefficients in (11). This is not an inference from (2); it uses the fixed Hodge structure of the smoothing.

We use the following complete results of van Geemen. For a general abelian fourfold of Weil type (A, K), the rational codimension-two Hodge classes form the three-dimensional fixed space $B_K^2 = \mathbf{Q}\omega_K^2 \oplus W_K$, and this space is constant in the Weil family under the natural identification of cohomology. For a general abelian surface J, one has $B^2(J \times J) = \operatorname{Sym}^2 B^1(J \times J)$, of dimension six; for the embedding of $K = \mathbf{Q}(\sqrt{-3})$ displayed in Lemma 1, the two Weil eigenclasses and the Weil polarization are the explicit forms below. These are `paper_id=van_Geemen_Weil_Classes_and_Decomposable_Abelian_Fourfolds`, `theorem_id=Definition2.1andSection2.3;Proposition3.1;Lemma3.3; Section6.2`, `arXiv=2108.02087`.

Put

$$a = \theta_1, \qquad b = \theta_2, \qquad c = m^*\theta - a - b,$$

where $m : J \times J \to J$ is addition, and let $\zeta = (-1 + \sqrt{-3})/2$. Then

$$\omega_+ = \zeta^2 a + \zeta c + b, \quad \omega_- = \zeta a + \zeta^2 c + b, \quad \omega_K = 2a - c + 2b. \tag{12}$$

Writing

$$W_s = \omega_+^2 + \omega_-^2, \qquad W_d = \frac{\omega_+^2 - \omega_-^2}{\zeta - \zeta^2},$$

the three forms ω_K^2, W_s, W_d are a basis of B_K^2.

We justify that a Hodge-generic choice is available in the Schoen smoothing, rather than assuming it. The local moduli space of Schoen surfaces is smooth and irreducible of dimension four by the smoothing theorem quoted in Lemma 1, and its Albanese varieties lie in the four-dimensional Weil family identified by van Geemen. The map to that Weil family is dominant. Indeed, for a smooth Schoen surface the rank-four alternating form making the Albanese embedding Lagrangian identifies

$$N_{S/A} \simeq \Omega_S^1.$$

Hence $h^0(N_{S/A}) = q(S) = 4$. Since a surface of general type has no global vector fields, the normal sequence gives an injection

$$H^0(A, T_A) \simeq \mathbf{C}^4 \hookrightarrow H^0(S, N_{S/A}),$$

which is therefore an isomorphism. Thus all first-order embedded deformations with the ambient abelian fourfold fixed are translations.

Here is the deformation-functor comparison. Add a full level structure and the polarization of type $(1, 1, 3, 3)$, so that locally the abelian moduli problem is fine, and let $\mathfrak{S}$ be the local moduli space of Albanese-embedded Schoen surfaces modulo translation. The Albanese universal property identifies the local abstract Schoen deformation functor with this embedded-pair functor: an abstract deformation has

its relative Albanese embedding, unique up to a relative translation. There is therefore a morphism

$$\Phi : \mathfrak{S} \longrightarrow \mathcal{W}_K$$

to the marked polarized Weil domain. For a fixed marked abelian fourfold, the tangent space of the Hilbert functor is $H^0(N_{S/A})$, while the tangent to the translation orbit is the image of $H^0(T_A)$. Consequently the kernel of $d\Phi$ is exactly

$$H^0(N_{S/A})/H^0(T_A) = 0. \tag{15a}$$

(The remaining automorphisms of a general marked polarized abelian variety are finite and have no tangent directions.) Thus Φ is generically quasi-finite. Both $\mathfrak{S}$ and the irreducible Weil domain $\mathcal{W}_K$ have dimension four, so the image of Φ has dimension four and is dense. This proves dominance with markings, polarization, and translation normalization all included.

The full-monodromy locus M^{full} constructed in Lemma 2 is nonempty Zariski open, while the points where the rational middle Hodge space is larger than B_K^2 form a countable union of proper Hodge loci in the Weil domain. By dominance their inverse images are still proper. Over the uncountable field $\mathbf{C}$, choose a complete-intersection curve through the Schoen boundary which is contained neither in the closed complement of M^{full} nor in any of those countably many proper loci. Its normalization gives a one-parameter smoothing whose very general point lies simultaneously in the full-monodromy locus and outside all extra Hodge loci. For this arc, the classes $s_2(\Lambda_\pm)$ are algebraic Hodge classes on every nonzero fibre; after the finite base change labeling them, their locally constant classes therefore lie in the fixed space B_K^2. Their specializations do as well.

The relevant surface classes are

$$\begin{gathered} [Y_+] = b(a+b-c), \qquad [Y_-] = a(a+b-c), \\ [\{0\}\times J] = \frac{a^2}{2}, \qquad [J\times\{0\}] = \frac{b^2}{2}, \qquad [\Delta_J] = \frac{(a+b-c)^2}{2}. \end{gathered} \tag{13}$$

Multiplication by two acts on surface homology by $2^4 = 16$. Hence (9)–(11) give, for nonnegative integers u_i, v_i,

$$\begin{aligned} \operatorname{sp} s_2(\Lambda_+) &= 16[Y_+] + u_1[\{0\}\times J] + u_2[J\times\{0\}] + u_3[\Delta_J], \\ \operatorname{sp} s_2(\Lambda_-) &= 16[Y_-] + v_1[\{0\}\times J] + v_2[J\times\{0\}] + v_3[\Delta_J]. \end{aligned} \tag{14}$$

For transparency, the coefficient comparison is included. Using $\zeta^2+\zeta+1=0$, expansion of (12) gives

$$\begin{aligned} W_s &= -a^2 - 2ab + 4ac + 2b^2 - 2bc - c^2, \\ W_d &= a^2 - 2ab + 2bc - c^2. \end{aligned} \tag{15}$$

The six monomials $a^2, b^2, c^2, ab, ac, bc$ are a basis of $B^2(J\times J)$. Requiring the first line of (14) to belong to the span of ω_K^2, W_s, W_d, and comparing these six coefficients, has the unique solution

$$(u_1,u_2,u_3) = (16,0,0), \qquad 16([Y_+] + [\{0\}\times J]) = \frac{8}{3}(\omega_K^2 + W_s). \tag{16}$$

The second line similarly has the unique solution

$$(v_1,v_2,v_3) = (0,16,0), \qquad 16([Y_-] + [J\times\{0\}]) = \frac{8}{3}\omega_K^2 - \frac{4}{3}W_s + 4W_d. \tag{17}$$

Thus the possible Δ_J-excess has multiplicity zero, while the two coordinate excess terms have exactly the multiplicities shown in (1). Equations (16)–(17) prove (1). □

Lemma 4. *For the smoothing in Lemma 3,*

$$\mathrm{sp}\big(s_2(\Lambda_+) - s_2(\Lambda_-)\big) = 8(b-a)(a+b-2c) \neq 0. \tag{18}$$

Proof. Subtracting the two identities in (1) and using (13) gives

$$16\left(b(a+b-c) + \frac{a^2}{2} - a(a+b-c) - \frac{b^2}{2}\right) = 8(b-a)(a+b-2c).$$

This class is nonzero. Indeed, on $\{0\} \times J$ one has $a = 0$, $c = 0$, and $b = \theta$, so the restriction of the unscaled class $\frac{1}{2}(b-a)(a+b-2c)$ is $\theta^2/2$, whose integral is one for a principal polarization on an abelian surface. Equivalently,

$$\int_{J\times J} 8(b-a)(a+b-2c)\, a^2 = 32 \neq 0.$$

□

Theorem 5. ***Conjecture: half-spin symmetry for conormal cycles***

Prove or disprove the following conjecture.

Let A be a complex abelian variety of dimension n, and let $Z \subset A$ be an irreducible, non-degenerate subvariety of dimension r. After translating Z if necessary, assume that Z is symmetric, namely $-Z = Z$.

Let $\Lambda_Z \subset T^*A$ be the conormal variety of Z, and let $\mathbf{P}\Lambda_Z \subset \mathbf{P}T^*A$ be its projectivization. Consider the projectivized conormal Gauss map

$$\gamma_Z : \mathbf{P}\Lambda_Z \longrightarrow \mathbf{P}(T_0^*A).$$

Assume that γ_Z is generically finite of degree $d = 2\ell$, where $\ell \geq 4$. Let $U \subset \mathbf{P}(T_0^*A)$ be a dense open subset over which γ_Z is finite etale. Assume that the monodromy group of $\gamma_Z|U$ is the even signed symmetric group

$$W(D_\ell) = \{\pm 1\}_+^\ell \rtimes \mathfrak{S}_\ell$$

in its natural action on the signed set $\{\pm 1, \ldots, \pm\ell\}$.

Since Z is symmetric, a general fiber may be written as

$$\gamma_Z^{-1}([\xi]) = \{\pm p_1, \ldots, \pm p_\ell\}.$$

For a sign vector $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_\ell) \in \{\pm 1\}^\ell$, define

$$p_\varepsilon = \sum_{j=1}^{\ell} \varepsilon_j p_j.$$

Let $\widetilde{U}_+$ and $\widetilde{U}_-$ be the two finite etale covers of U parametrizing the sign vectors satisfying, respectively,

$$\varepsilon_1 \cdots \varepsilon_\ell = +1$$

and

$$\varepsilon_1 \cdots \varepsilon_\ell = -1.$$

Equivalently, $\widetilde{U}_+$ parametrizes sign choices with an even number of minus signs, while $\widetilde{U}_-$ parametrizes those with an odd number.

Apply the summation map

$$(\varepsilon, [\xi]) \longmapsto (p_\varepsilon, [\xi]) \in A \times \mathbf{P}(T_0^* A) = \mathbf{P}T^*A.$$

Let $\mathbf{P}\Lambda_+$ *and* $\mathbf{P}\Lambda_-$ *be the cycle-theoretic closures of the two images, with the generic degrees of the summation maps retained as multiplicities. Let* Λ_+ *and* Λ_- *denote the corresponding conic cycles in* T^*A.

Then Λ_+ *and* Λ_- *are effective irreducible conic Lagrangian cycles, each having Gauss degree* $2^{\ell-1}$. *They correspond to the two Weyl-group orbits of weights of the half-spin representations* $\mathbb{S}^+$ *and* $\mathbb{S}^-$ *of* $\mathrm{Spin}(2\ell)$. *The unordered pair* $\{\Lambda_+, \Lambda_-\}$ *is independent of the signed labeling of the general Gauss fiber.*

Let $\pi_A : \mathbf{P}T^*A \to A$ *be the natural projection, and put*

$$X_\pm = \pi_A(\mathrm{Supp}\, \mathbf{P}\Lambda_\pm).$$

The conjecture is that:

(1) The two projected supports have the same dimension:

$$\dim X_+ = \dim X_-.$$

(2) The two projectivized Lagrangian cycles have the same integral homology class:

$$[\mathbf{P}\Lambda_+] = [\mathbf{P}\Lambda_-]$$

in $H_{2n-2}(\mathbf{P}T^*A, \mathbf{Z})$.

Equivalently, if $h = c_1(\mathcal{O}_{\mathbf{P}T^*A}(1))$ *and*

$$s_i(\Lambda_\pm) = (\pi_A)_*(h^{n-1-i} \cap [\mathbf{P}\Lambda_\pm]) \in H_{2i}(A, \mathbf{Z}),$$

then the second assertion states that

$$s_i(\Lambda_+) = s_i(\Lambda_-)$$

for every $0 \le i \le n-1$.

Remark. When ℓ *is odd, both assertions hold because inversion exchanges the two half-spin cycles:* $\mathbf{P}\Lambda_- = (-1)_*\mathbf{P}\Lambda_+$. *When* $\ell = 2k$ *is even, it is known that* $s_i(\Lambda_+) = s_i(\Lambda_-)$ *for* $0 \le i \le k-1$, *and that* $\dim X_+ = \dim X_-$ *whenever* $k - 1 > \min\{\dim X_+, \dim X_-\}$. *Thus, the genuinely conjectural case is* $\ell = 2k$ *with* $\min\{\dim X_+, \dim X_-\} \ge k-1$*; the first possible difference between the homology classes occurs in the component* s_k. *No elementary counterexample satisfying all the hypotheses is presently apparent.*

Proof. The conjecture is false. Take a very general Schoen surface $Z = S_t \subset A_t$ from Lemma 1, on an arc chosen as in Lemma 3. It is smooth, irreducible, symmetric after translation, and Albanese embedded in an abelian fourfold. Its conormal Gauss map has degree eight, hence $\ell = 4$, and is finite over a dense open set; finiteness of its Gauss map also implies non-degeneracy. By Lemma 2, its exact natural monodromy is $W(D_4)$. Thus every hypothesis in the question is satisfied.

After the harmless finite base change labeling the two parity cycles, intersection with the relative class h, proper pushforward to the abelian scheme, and specialization commute. If the two projectivized half-spin cycles on the general fibre had the same integral homology class, then in particular their s_2-classes would agree, and their difference would specialize to zero. Lemma 4 instead gives

$$\mathrm{sp}\big(s_2(\Lambda_+) - s_2(\Lambda_-)\big) = 8(b-a)(a+b-2c) \neq 0.$$

Therefore $[\mathbf{P}\Lambda_+] \neq [\mathbf{P}\Lambda_-]$ already in rational, and hence also in integral, homology. This Schoen surface is a counterexample to assertion 2, so the stated conjunction is disproved. □

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